Mohammad Khan

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Experience

Saronic Technologies July 2026 - Present
Perception Engineer *Austin, TX*

- Build and maintain Databricks pipelines that prepare sensor data for training and evaluating perception models on autonomous surface vessels
- Develop internal data tools that perception and autonomy engineers use to prepare and analyze datasets
- Take part in on-water field testing of autonomous vessels and use test data to evaluate perception performance

Naval Surface Warfare Center, Crane Division Summers 2021 - 2023
X-Force Fellow *Crane, IN*

- Built a virtualized development workflow for autonomous drones that improved development efficiency by 60%
- Wrote basic autonomous navigation algorithms for quadcopters and an IR beacon precision landing system for landings in non-traditional environments
- Built ground robots and robotic arms for a simulated IED-defusing task (2021, 2022)

Education

University of Texas at Dallas Aug 2022 - May 2026
B.S. Neuroscience (Major Honors) and B.S. Mathematics, Statistics Concentration *Richardson, TX*

- Minor in Computer Science; Cum Laude; GPA: 3.86/4.00

Research

Unsupervised Behavior Clustering Feb 2024 - May 2026
Neurobiology of Basal Ganglia Function and Dysfunction Lab, UT Dallas *Richardson, TX*

- Used unsupervised clustering in MATLAB and Python on wireless accelerometer data to identify behavioral syllables in mouse models of degenerative brain disease
- Designed and built a 3D arena for recording naturalistic mouse behavior across several sensing modalities
- Presented an award-winning poster at the UT Dallas Summer Platform for Undergraduate Research (SPUR) Symposium, July 2024

Projects

Robosub, UT Dallas Aug 2023 - May 2026
MATE ROV Programming Lead, Roboboat Programming Member *Richardson, TX*

- Took over as programming lead after the original lead left and worked with the other team leads to get the ROV competition ready; the team placed 3rd at the 2024 MATE ROV World Championships in its first appearance
- Wrote the ROV control software from scratch in ROS2, Python, and Docker, including PID depth hold and the network link that carries video and commands between the operator interface and the vehicle
- Added Jetson Orin companion computers to an autonomous surface vessel's Cube Orange autopilot to run LiDAR, GPS, and computer vision processing

Comparative Transcriptomic Analysis of Brain Injury Jan 2025 - May 2025
R, Seurat *UT Dallas*

- Compared gene expression in mouse brain cells under control and injury conditions using bulk and single-cell RNA-seq data
- Preprocessed, normalized, and clustered the data in R with Seurat, then ran differential expression analysis
- Mapped expression changes across brain regions with spatial transcriptomics and identified cell-type-specific responses to injury

Skills

Languages: Python, C++, MATLAB, R
Robotics: ROS2, ArduPilot, MAVROS, Gazebo, PID control, I2C
Bioinformatics: Seurat, bulk and single-cell RNA-seq, spatial transcriptomics, differential expression
Data and Tools: Databricks, Docker, Git, Linux, Anaconda
Certifications: Nationally Registered EMT (NREMT)